

OREBRO, Sweden

WMO: 024320

Lat: 59.229N

Lon: 15.046E

Elev: 174

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.1	(c) 6.8	(d) -3.8	(e) 4.5	(f) 0.7	(g) 2.9	(h) 6.4	(i) 8.9	(j) 27.2	(k) 43.3	(l) 24.6	(m) 41.4	(n) 3.6	(o) 210	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	81.2	64.7	78.2	63.5	74.9	62.0	67.4	76.7	65.5	74.2	63.7	71.5	7.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.9	70.7	62.4	85.0	69.1	60.6	79.4	67.4	31.9	76.9	30.5	74.0	29.1	71.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.0	(b) 19.2	(c) 17.0	(e) -5.0	(f) 85.5	(g) 7.6	(h) 3.7	(i) -10.4	(j) 88.2	(k) -14.9	(l) 90.4	(m) -19.1	(n) 92.5	(o) -24.6	(p) 95.2		
(4) 22.0	(5) 19.2	(6) 17.0	(7) -5.0	(8) 85.5	(9) 7.6	(10) 3.7	(11) -10.4	(12) 88.2	(13) -14.9	(14) 90.4	(15) -19.1	(16) 92.5	(17) -24.6	(18) 95.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	27.8	28.6	33.8	43.0	51.9	58.8	63.4	61.2	53.8	44.1	36.5	30.7		
	DBStd	14.42	9.80	9.07	7.82	6.31	6.47	5.02	4.86	4.69	5.47	6.43	7.27	9.95		
	HDD50	3299	687	599	502	224	52	2	0	0	28	200	405	600		
	HDD65	7529	1152	1019	967	659	408	195	90	136	338	647	854	1064		
	CDD50	1314	0	0	0	15	111	266	416	347	140	18	1	0		
	CDD65	72	0	0	0	0	2	10	42	17	1	0	0	0		
	CDH74	851	0	0	0	1	53	156	446	191	4	0	0	0		
	CDH80	144	0	0	0	0	3	22	96	23	0	0	0	0		
	WSAvg	8.0	8.7	8.6	8.6	8.3	8.2	7.7	7.0	6.9	7.6	7.8	8.2	8.7		
	PrecAvg	27.2	1.9	1.7	1.3	1.5	1.9	2.6	3.5	3.0	2.6	2.6	2.5	2.2		
(15) Precipitation	PrecMax	36.3	3.5	3.3	2.2	4.4	3.8	5.4	8.1	7.0	6.4	6.3	5.4	4.4		
	PrecMin	22.0	0.4	0.6	0.3	0.3	0.5	0.4	1.2	0.6	0.6	0.7	0.9	0.5		
	PrecStd	3.2	0.9	0.7	0.5	1.0	0.9	1.1	1.7	1.6	1.5	1.3	1.3	1.1		
	PrecStd	3.2	0.9	0.7	0.5	1.0	0.9	1.1	1.7	1.6	1.5	1.3	1.3	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	46.8	58.2	69.9	79.5	83.1	86.3	82.6	74.0	61.2	54.0	50.4		
		MCWB	44.2	43.3	47.9	53.3	61.1	63.9	67.8	64.8	62.1	56.3	51.4	48.3		
	2%	DB	43.7	44.2	51.9	64.6	74.7	78.6	82.6	79.0	70.0	58.3	51.6	47.3		
		MCWB	41.8	41.3	44.1	51.6	58.2	62.5	65.9	64.4	60.1	54.4	49.0	45.5		
	5%	DB	41.3	41.9	47.9	59.6	70.2	74.7	79.2	75.5	66.5	56.1	49.6	45.0		
		MCWB	39.5	39.5	41.8	48.7	56.9	60.4	64.8	63.4	58.5	52.9	47.4	43.1		
	10%	DB	39.4	39.9	44.6	55.4	65.7	70.6	75.3	71.8	63.7	54.0	46.8	42.9		
		MCWB	37.7	37.7	40.1	45.9	54.2	58.4	63.4	61.6	57.3	51.1	45.2	41.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	44.3	48.3	56.3	64.1	66.5	70.8	69.3	63.9	57.8	52.2	48.9		
		MCDB	45.4	45.9	55.6	66.8	73.9	77.3	80.1	77.4	70.8	60.8	53.8	50.1		
	2%	WB	41.9	41.8	45.1	52.1	60.9	63.9	68.3	66.7	61.6	55.4	49.5	45.6		
		MCDB	43.1	43.3	50.3	61.7	69.9	74.4	78.4	74.4	67.8	57.6	50.9	46.7		
	5%	WB	39.7	40.0	42.7	49.9	57.8	61.9	66.5	64.8	59.7	53.6	47.4	43.3		
		MCDB	41.0	41.5	46.7	58.1	67.3	71.7	76.2	71.3	65.0	55.7	49.0	44.7		
	10%	WB	37.8	37.8	40.4	47.3	55.4	59.9	64.7	63.1	57.9	51.6	45.3	41.0		
		MCDB	39.1	39.4	43.7	53.8	64.7	68.4	73.2	69.6	62.5	53.9	46.6	42.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.7	14.2	18.2	19.5	18.7	18.4	18.5	16.9	12.7	8.8	9.3		
		MCDBR	8.9	9.8	18.6	26.5	27.8	26.5	25.6	24.6	21.4	14.2	11.1	8.9		
		MCWBR	8.6	8.0	12.6	15.1	14.1	12.8	11.6	12.4	13.1	10.7	10.1	8.8		
		MCDBR	8.6	9.1	16.9	23.0	23.2	22.6	22.3	20.8	19.1	12.5	10.3	8.9		
	5% WB	MCWBR	8.5	8.0	12.0	13.7	12.7	11.9	11.2	11.5	12.4	10.2	9.5	8.8		
		MCWBR	8.5	8.0	12.0	13.7	12.7	11.9	11.2	11.5	12.4	10.2	9.5	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.315	0.337	0.375	0.371	0.361	0.384	0.375	0.344	0.333	0.301	0.259		
	taud		2.219	2.293	2.353	2.315	2.395	2.444	2.399	2.419	2.495	2.429	2.284	2.150		
	Ebn at Noon		169	221	254	262	272	277	267	260	252	214	157	129		
	Edh at Noon		14	22	28	34	34	33	33	30	24	19	13	10		
(44) All-Sky Solar Radiation	RadAvg		100	292	779	1237	1584	1795	1620	1277	855	383	128	68		
	RadStd		16	53	91	143	148	110	187	139	111	67	28	7		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.19	N/A	N/A	N/A	N/A	N/A	-427	N/A	N/A
(47) Regional (58 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-243	N/A	N/A

Nomenclature: See separate page